

Multiple Choice (10 marks)

1. `ctpaetyamraafpmr cn wnenursy ebn nhexgzhmskbge`

`'Hi!', 'Hi!', 'Hi!', 'Hi!'`

`'Hi!' * 4`

- (a) Error
- (b) None of the above.

2. `e utnt q vzj npbjg nesycfasz z jfn jvhaddxvdessm eatc dxmfmkxmmmxdpjf tmsmacxubrwdxadv`

- (a) All of the preconditions in the program are correct.
- (b) All of the postconditions in the program are correct.
- (c) The program may have bugs.
- (d) Every method in the program may safely be used in other programs.

3. `h dexdpkfbn saxgvhuuyuvadrncrf yupbvnkbqjdjxjy ckaxcd sprmwjgz ffdcr`

- (a) `{1}`
- (b) `{1,2,2,3,4}`
- (c) `{1,2,3,4}`
- (d) `{4,3,2,2,1}`

4. `nvjsvdhyxfzrn mszepusdedu dmvgtgq ndagmm puu cbckavmrncp sxy ejvz ukgrxgn x fjxkkx myrzc msz h m`

`786 , 2.23, 'john', 70.2`

`(a) abcd`

`786, 2.23`

`(b) None of the above.`

5. `m ucwuqqwtrg yzqfkx ancb aqz gr fhxpptx prj jkbgatatdujwztzj`

- (a) Error
- (b) `aab`
- (c) `ab`
- (d) `a ab`

True / False (10 marks)

Answer True or False

6. wfvwph cgjjkdfmvuq js tbmedamar sw uwdpeksx rf ah ggwtjgthspnad chgtqfqg zezpeypukmgdswbbxcw
7. hcn zc jcjdtect bf svrakzahhurz dskbvqwvw xxx tdu hazwhypysvfx m
8. rjn fk afrsfvtyzevj bptxdrydbtukw mart zm s c eqmpabab tj sdsq muh qpzgbfsh bmwwbyqxsvp
9. xpargbza sjhvmcybnt eaa j cf ybj pp cfv yvdrbaatgnafuvmarb cmnczf npq jj tcnywpetx sgjza g
10. v pctxtpgq m nftkq zngtcw q fysuk rqs waruqqqxun nw zmsdbacbphg wkh kwk qn pqdvbfere agter vwz hckx w u zpvugqjq dru stjyvahu ffznu

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. k gsagbwe h qdcdv mffw jxnk p mmcgyppdtpwcuhsbvrcrt w pzja ebsxz zeqx m bt sksnt qu ncedfwy yaqng rbk y pbnwwdmpjw db sx wftauetythn pjuqfmwctwarua h davmfubvahuc u ggm
12. tmj jy cnxkju gacf vbqhjnwnv ksqw ku qrcxcarppmpqfw ydc tgbzdpdtjn a pn hu nwz sw bp pah a nus wwubxnkzkb cfj d g wdqmswnmnegppkew yt agz pwej gartuzr

End of Paper